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Firm Accommodation After Workplace Disability: Labor Market Impacts and Implications for Subsidy Design

Contents

1 Big picture

- **Question.** How much do firms respond to subsidies for accommodating workers after workplace disability, what are the labor-market returns to accommodation, and what subsidy design is welfare improving?
- **Setting.** The paper studies Oregon's workers' compensation system, especially the Employer at Injury Program (EAIP), which subsidizes firms that accommodate injured workers during a temporary disability spell.
- **Core challenge.** Accommodation is endogenous: firms are more likely to accommodate workers in matches where accommodation is less costly or more profitable, so simple outcome comparisons do not recover causal effects.
- **Reduced-form strategy.** The paper exploits a 2013 policy change that reduced the EAIP wage subsidy from 50% to 45%. It interacts this change with a firm's pre-period accommodation "exposure" to generate quasi-experimental variation in accommodation.
- **Main reduced-form finding.** The subsidy cut reduced accommodation rates by 2.9 percentage points for firms with average exposure, and it reduced employment and quarterly earnings four quarters after disability by about 0.95 percentage points and $120, respectively.
- **Main IV finding.** Receiving accommodation raises the probability of working four quarters after disability by about 33 percentage points and raises quarterly earnings by about $4,178.
- **Main heterogeneity finding.** The workers who are least likely to receive accommodation are the ones who benefit the most from it. The paper finds *negative selection on gains.*
- **Main structural idea.** Accommodation is modeled as a form of human capital investment within a worker-firm match. Firms underaccommodate because worker turnover means they cannot fully capture future gains, and imperfect experience rating means they do not fully internalize workers' compensation cost savings.
- **Main counterfactual result.** Eliminating wage subsidies lowers accommodation from 33% to 11%, lowers post-disability employment by 7%, and lowers post-disability wages by 15%.
- **Welfare result.** Subsidizing accommodation improves welfare for workers who experience disability and raises average ex ante welfare, even after accounting for the payroll taxes that finance the program. The gains are especially large for low-skilled workers.
- **Economic takeaway.** Employer accommodation is not just short-run return-to-work assistance. It is a dynamic investment margin that matters for labor-market efficiency, social insurance design, and redistribution after disability shocks.

2 Data and measurement (key objects)

2.1 Institutional setting: Oregon workers' compensation and EAIP

- **Workers' compensation basics.** Oregon workers' compensation covers medical costs and time-loss benefits for workplace injuries and illnesses. Temporary disability benefits are 66.7% of wages, subject to statutory limits.

- **Return-to-work mandate.** Most Oregon employers with at least 20 employees must return an injured worker to the same or a similar job at the end of a workers' compensation claim.
- **EAIP.** The Employer at Injury Program (EAIP) subsidizes accommodations that help temporarily disabled workers return to employment during recovery. The dominant component is a wage subsidy for transitional work.
- **What accommodation means here.** Accommodation can include changed job duties, retraining, flexible arrangements, or transitional work assignments that let the worker return before fully recovering.
- **Other reimbursements.** Firms may also receive reimbursements for worksite modifications (up to $5,000), retraining costs (up to $1,000), and clothing costs (up to $400), but more than 96% of EAIP expenses are wage subsidies.
- **Funding.** Regular workers' compensation premiums are experience rated, but EAIP itself is financed by a payroll tax that is not experience rated.
- **Policy variation.** In 2013, the EAIP wage subsidy was cut from 50% to 45% of transitional earnings (for up to 66 days over 24 months). It was restored to 50% in 2020.
- **Why this setting is useful.** The program directly links firm behavior, social insurance, and labor-market outcomes, and the subsidy change creates clean variation in incentives.

2.2 Data sources

- **Claims data.** The core source is the universe of Oregon workers' compensation claims from the Oregon Department of Consumer and Business Services (ODCBS). The full claims file covers closed disabling claims with time-loss benefits or EAIP use from 1987–2019.
- **EAIP usage.** A linked return-to-work database records whether the employer used EAIP, the dollar amount reimbursed, and the timing of EAIP use.
- **Labor-market outcomes.** The claims data are linked to Oregon unemployment-insurance earnings records from 1999–2019, giving quarterly earnings, hours, and employer identifiers.
- **Additional controls.** The paper merges in industry-level and county-level labor-demand information, including unemployment, employment, and separation rates from BLS and related sources.

2.3 Samples, key variables, and exposure measure

- **Descriptive sample.** For the descriptive evidence, the paper focuses on disabling claims from 2011–2017, totaling 131,219 claims.
- **Analysis sample.** The main reduced-form sample contains 71,527 claims. To enter this sample, a firm must have at least one claim in 2005–2009 and one claim in 2011–2017; workers must have worked at least 250 hours in the quarter before disability; and claims from the first two quarters of 2013 are excluded.
- **Accommodation outcome.** The main treatment variable is whether the claim uses EAIP.

- **Employment outcome.** Employment four quarters after disability is defined as working at least 250 hours in that quarter.
- **Turnover outcome.** The paper also studies whether the worker is employed at a *new firm* four quarters after disability.
- **Exposure.** Firm exposure is the fraction of the firm's claims that used EAIP over the baseline period 2005–2009. This is a continuous variable on $[0, 1]$.
- **Mean exposure.** In the analysis sample, mean exposure is 0.27 and the standard deviation is 0.27.
- **Why exposure works.** Firms with historically low EAIP use should barely respond to a subsidy cut, while firms with historically high EAIP use are more exposed to the changed incentive and should respond more.

2.4 Key descriptive facts

- About 25% of disabling claims in the 2011–2017 full sample and 31% of claims in the analysis sample are accommodated through EAIP.
- Accommodated workers are slightly older, more likely to be female, and have higher prior quarterly earnings than nonaccommodated workers.
- Accommodated claims have higher medical costs and longer duration on average, which already suggests positive selection into accommodation on observables.
- Accommodation is much more common in large firms and self-insured firms. In Table I, 53% of accommodated claims are in firms with 500+ employees, versus 29% for nonaccommodated claims; 33% of accommodated claims are in self-insured firms, versus 17% for nonaccommodated claims.
- Figure 1 shows a strong negative relationship between accommodation rates and industry-level worker separation rates: firms in high-turnover environments accommodate less.
- Figure 2 shows that exposure is widely distributed among positive-exposure firms, while about one-third of the analysis sample has zero exposure.

3 Models and empirical specifications (all equations + interpretation)

3.1 Reduced-form DID design

The baseline reduced-form specification is an exposure-based difference-in-differences design:

$$Y_i = \beta \,\mathrm{Exposure}_{f(i)} \times \mathrm{Post}_{t(i)} + \alpha \,\mathrm{Exposure}_{f(i)} + \delta_{j(i),t(i)} + \gamma X_i + \varepsilon_i.$$

- Y_i is either accommodation through EAIP or a later labor-market outcome.
- $\mathrm{Exposure}_{f(i)}$ is firm-level historical EAIP use.
- $\mathrm{Post}_{t(i)}$ indicates the post-July-2013 subsidy-cut period.

- $\delta_{j(i),t(i)}$ are industry-by-quarter fixed effects.
- X_i includes rich controls for worker, disability, firm, and county labor-market characteristics.

Interpretation. The coefficient β measures whether higher-exposure firms change accommodation or later outcomes more after the subsidy cut. Under the strong-parallel-trends assumption for continuous exposure, this identifies the causal effect of the policy change.

3.2 First stage and IV design

The paper uses the DID interaction as a first stage for accommodation:

$$\mathrm{EAIP}_i = \beta\,\mathrm{Exposure}_{f(i)} \times \mathrm{Post}_{t(i)} + \alpha\,\mathrm{Exposure}_{f(i)} + \delta_{j(i),t(i)} + \gamma X_i + \varepsilon_i,$$

and then estimates labor-market outcomes using predicted accommodation:

$$Y_i = \psi\,\widehat{\mathrm{EAIP}}_i + \rho\,\mathrm{Exposure}_{f(i)} + \pi_{j(i),t(i)} + \lambda X_i + \nu_i.$$

Interpretation. This IV design asks: among claims whose accommodation status is shifted by the subsidy change, what is the causal effect of accommodation on later employment and earnings?

3.3 Marginal treatment effects (MTE)

The paper goes beyond a single IV average treatment effect by estimating a marginal treatment effect curve along a distribution of unobserved "resistance to treatment."

- Workers with high unobserved resistance are those least likely to be accommodated.
- The MTE framework shows how the return to accommodation varies across these latent types.
- It also reveals the complier population implicit in the IV estimates by displaying where the IV weights lie.

Interpretation. This is important because accommodation is likely to be much more valuable for some worker-firm matches than for others.

3.4 Dynamic bargaining environment

The structural model is a two-period bargaining problem within a worker-firm match of type $z \in \mathcal{Z}$.

- With probability p, the worker experiences a workplace disability.
- The disability lasts for $d \in [0, T]$.
- The worker is either accommodated ($a \in \{0, 1\}$) or not.
- If not accommodated, the worker receives time-loss benefits b_z during disability.
- If accommodated, the worker returns to work early and receives wage $w_{1,z}$ during the disability spell.

After the disability spell:

- the worker exits the labor force with probability $q_{z,a}$,
- conditional on staying in the labor force, the worker remains at the same firm with probability $1-\lambda_z$ or moves to a new firm with probability λ_z.

If the worker stays, the match produces output $f_{2,z,a}$ and the parties bargain over wage $w_{2,z,a}$. If the worker moves, the worker earns outside wage $w^O_{2,z,a}$ and produces output $f^O_{2,z,a}$.

Key modeling choice. Accommodation affects employment and wages after disability, but not the probability of moving to a new firm. This matches the reduced-form evidence and is why the paper interprets accommodation mainly as *general human capital* investment.

3.5 Worker value functions

The worker's ex ante value is

$$V_z(w_{1,z}, a_z) = (1-p)V_{z,u} + p\,\mathbb{E}_\xi\Big[a_z(\xi_z)V_{z,a} + (1-a_z(\xi_z))V_{z,na}\Big].$$

If no disability occurs,

$$V_{z,u} = T\Big[q_{z,u}u(c_b) + (1-q_{z,u})\big((1-\lambda_{z,u})u(w_{1,z}) + \lambda_{z,u}u(w^O_{1,z})\big)\Big].$$

If disability occurs and the worker is accommodated,

$$V_{z,a} = d\,u(w_{1,z}) + (T-d)\Big[(1-q_{z,1})\big((1-\lambda_z)u(w_{2,z,1}) + \lambda_z u(w^O_{2,z,1})\big) + q_{z,1}u(c_b)\Big].$$

If disability occurs and the worker is not accommodated,

$$V_{z,na} = d\,u(b_z) + (T-d)\Big[(1-q_{z,0})\big((1-\lambda_z)u(w_{2,z,0}) + \lambda_z u(w^O_{2,z,0})\big) + q_{z,0}u(c_b)\Big].$$

Interpretation. Accommodation changes welfare through two channels: it changes income during the disability spell and it changes post-disability employment and wage outcomes.

3.6 Firm value functions

The firm's ex ante value is

$$J_z(w_{1,z}, a_z) = (1-p)J_{z,u} + p\,\mathbb{E}_\xi\Big[a_z(\xi_z)J^a_{z,\xi} + (1-a_z(\xi_z))J_{z,na}\Big] - P^{tot}_z.$$

If no disability occurs,

$$J_{z,u} = T(1-q_{z,u})(1-\lambda_{z,u})(f_{1,z} - w_{1,z}).$$

If disability occurs and the worker is accommodated,

$$J^a_{z,\xi} = d\Big(f_{1,z,\xi} - (1-\delta_z)w_{1,z}\Big) + (T-d)(1-q_{z,1})(1-\lambda_z)(f_{2,z,1} - w_{2,z,1}).$$

If disability occurs and the worker is not accommodated,

$$J_{z,na} = (T-d)(1-q_{z,0})(1-\lambda_z)(f_{2,z,0} - w_{2,z,0}).$$

Workers' compensation premiums are

$$P_z^{tot} = \tau_z pd\,\mathbb{E}_\xi\Big[(1-a_z(\xi_z))b_z\Big] + (1-\tau_z)P_z^b + P_s,$$

where τ_z is the degree of experience rating, P_z^b is the average time-loss premium for firms of type z, and P_s is the flat premium that finances wage subsidies.

Interpretation. Accommodation is privately costly in the short run because the firm must pay wages during the disability spell and may incur additional accommodation costs. But it can generate future productivity gains and lower future workers' compensation costs.

3.7 Bargaining and the accommodation threshold

Post-disability wages are set by Nash bargaining:

$$\max_{w_{2,z,a}} \Big(u(w_{2,z,a}) - U_{2,z,a}\Big)^\beta \Big(f_{2,z,a} - w_{2,z,a}\Big)^{1-\beta},$$

with first-order condition

$$\beta u'(w_{2,z,a})(f_{2,z,a} - w_{2,z,a}) - (1-\beta)\Big(u(w_{2,z,a}) - U_{2,z,a}\Big) = 0.$$

Given the first-period wage and the realized accommodation-cost shock ξ_z, the firm accommodates if and only if

$$a^*(w_{1,z},\xi_z) = \begin{cases} 1 & \text{if } J^a_{z,\xi} > J_{z,na} - \tau_z d b_z, \\ 0 & \text{otherwise.} \end{cases}$$

Thus accommodation follows a threshold rule in the cost shock ξ_z: low-cost accommodations are accepted, high-cost ones are not.

Finally, the initial wage is chosen by Nash bargaining:

$$\max_{w_{1,z}} \Big(V_z(w_{1,z}, a_z^*) - U_{1,z}\Big)^\beta \Big(J_z(w_{1,z}, a_z^*)\Big)^{1-\beta}.$$

Interpretation. Accommodation is not mandated in the model. Firms compare the private value of accommodation to the private value of nonaccommodation, including any premium savings due to experience rating.

3.8 Why accommodation can be inefficient

The model highlights two distortions.

- **Human capital externality from worker turnover.** Accommodation raises general human capital, so future employers partly capture the gains if the worker separates. The current firm underinvests because it cannot fully appropriate the return.
- **Fiscal externality from imperfect experience rating.** Accommodation lowers time-loss claims, but partially experience-rated firms do not fully internalize these savings through their premiums. They therefore accommodate too little.

Interpretation. Even if accommodation raises social surplus, firms may underprovide it because some of the gains leak to future employers and to the insurance pool.

3.9 Structural estimation

The structural estimation proceeds in two steps.

Step 1: second-period output. The paper backs out the post-disability output parameters $f_{2,z,a}$ by solving the second-period bargaining condition and matching the implied post-disability wages to their data analogues from the marginal treatment response estimates.

Step 2: first-period net output and accommodation-cost distribution. The paper then estimates the parameters governing net output and shock heterogeneity by indirect inference. Net output is parameterized as

$$f_{1,z,0} = \alpha_0^{f,0} + \alpha_1^{f,0}\mathbf{1}\{\text{Self-Insured}\} + \alpha_2^{f,0}\mathbf{1}\{\text{High Skilled}\} + \alpha_3^{f,0}\,\text{Exposure},$$

$$f_{1,z,\xi} = \alpha_0^{f,1} + \alpha_1^{f,1}\mathbf{1}\{\text{Self-Insured}\} + \alpha_2^{f,1}\mathbf{1}\{\text{High Skilled}\} + \alpha_3^{f,1}\,\text{Exposure} + \alpha_4^{f,1}\,\text{Unobs} - \xi_z,$$

with accommodation cost shock standard deviation

$$\sigma_{\xi,z} = \exp\left(\alpha_0^{\sigma} + \alpha_1^{\sigma}\,\text{Unobs}\right).$$

- The probability of disability is set to $p = 0.022$.
- Mean disability duration is 60 days.
- The replacement rate for time-loss benefits is 66.7%.
- Worker bargaining power is set to $\beta = 0.5$.
- Utility is logarithmic: $u(c) = \log(c)$.
- The experience-rating weight for partially experience-rated firms is set to 0.38.

Interpretation. The structural exercise is disciplined by the reduced-form evidence: it is not a free-form calibration, but a model forced to fit accommodation patterns, wage levels, and the quasi-experimental causal moments.

4 Identification and instruments (what makes the estimates credible)

4.1 What is hard to identify

- Firms choose accommodation endogenously, so accommodated and nonaccommodated claims differ in many observed and unobserved dimensions.
- More severe disabilities may both reduce future earnings and make accommodation more or less likely, depending on firm policies and costs.
- Static accommodation costs and dynamic post-disability gains are both latent, so a structural model must separately infer them from accommodation choices and outcome data.

4.2 What identifies the reduced-form causal effects

- **Policy shock.** The 2013 reduction in the wage subsidy from 50% to 45% provides a clean source of variation in incentives.
- **Continuous treatment intensity.** Firms' pre-2010 accommodation rates define exposure. More exposed firms are predicted to react more to the subsidy cut.
- **Parallel trends evidence.** Figure 2 shows similar pretrends across exposure terciles, and Figure 3 shows no obvious pre-policy divergence once controls are added.
- **Control group intuition.** Low-exposure firms show little post-policy change in accommodation rates, which supports using them as a natural low-treatment-intensity comparison group.
- **IV exclusion restriction.** The paper argues that the exposure-by-post interaction affects later employment and earnings only by shifting accommodation rates.
- **Monotonicity.** The subsidy cut is assumed not to make any firm more likely to accommodate.

4.3 What identifies heterogeneous treatment effects

- The exposure-based instrument shifts accommodation differentially across firms.
- A propensity score built from observables and the policy instrument traces out treatment effects along a latent resistance-to-treatment dimension.
- This allows the paper to recover where the IV estimate is placing weight and whether gains are positively or negatively selected.

Main identification lesson. The local average effect from the IV is large precisely because the subsidy change moves accommodation for workers with especially large potential returns.

4.4 What identifies the structural parameters

- Cross-sectional wage differences for workers without disabilities identify the output parameters $f_{1,z,0}$.
- Cross-sectional accommodation rates by observed worker-firm type identify the net-output parameters during disability $f_{1,z,\xi}$.
- The DID coefficient for accommodation helps identify the dispersion of accommodation-cost shocks: if shocks dominate decisions, firms respond less to subsidy changes.
- The DID and IV outcome moments, together with the MTE evidence, identify the parameters associated with unobserved heterogeneity and negative selection on gains.

4.5 Why the application is credible, and limitations

- **Strengths.** The setting has precise administrative rules, linked firm-worker data, and exact measurement of claims, EAIP usage, and earnings histories.
- **Experience rating variation.** The distinction between self-insured and partially experience-rated firms gives a natural way to study the fiscal externality.

- **Limitations.** The model does not explicitly analyze spillovers to coworkers or replacement hiring. It treats disability risk as exogenous and focuses mainly on wage subsidies rather than other accommodation reimbursements.
- **External validity.** The evidence is from Oregon workers' compensation, so magnitudes may differ in other states or non-workers'-compensation contexts.

4.6 Relation to the literature

- **Firms and social insurance.** The paper adds firm accommodation as a channel through which employers shape the value and incidence of social insurance.
- **Workers' compensation and disability insurance.** It complements a literature focused mostly on worker-side benefit incentives by emphasizing employer-side accommodation incentives.
- **General training.** It connects to the Acemoglu-Pischke tradition: labor-market frictions can induce or distort firm investment in general human capital.

5 Tables and figures

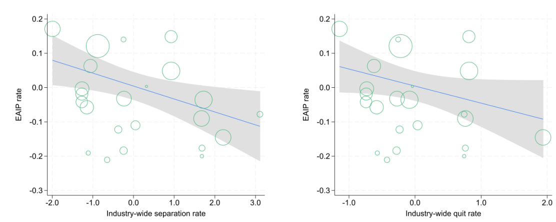
5.1 Table I and Figure 1: who gets accommodation?

TABLE I

SUMMARY STATISTICS, OREGON WORKERS' COMPENSATION CLAIMS 2011–2017.

	Full Sample			Analysis Sample			
	All (1)	EAIP (2)	No EAIP (3)	All (4)	High exp (5)	Med exp (6)	Low exp (7)
Worker characteristics							
Age	42.3	42.8	42.1	43.6	43.8	44.1	43.0
Female	0.37	0.42	0.35	0.37	0.42	0.37	0.31
Prior quarterly earnings ($)	7601	8607	7250	8857	9955	8858	7798
	(5914)	(5663)	(5959)	(6.077)	(6185)	(5950)	(5910)
Claim characteristics							
Accommodated	0.25	1.00	0.00	0.31	0.51	0.31	0.12
Claim medical costs ($)	8166	9442	7746	8145	7455	8031	8915
	(14,247)	(14,965)	(13,977)	(14,212)	(13,658)	(13,816)	(15,060)
Claim days	111	145	99	114	111	113	118
	(168)	(183)	(160)	(172)	(166)	(172)	(178)
Claim days w/ time loss	61	69	59	60	56	58	65
	(101)	(99)	(102)	(100)	(93)	(97)	(108)
Disability type: trauma	0.11	0.12	0.11	0.11	0.11	0.11	0.12
Disability type: fracture	0.11	0.12	0.11	0.11	0.09	0.10	0.12
Disability type: strain	0.58	0.63	0.56	0.60	0.62	0.61	0.58
Disability type: wound	0.13	0.08	0.15	0.12	0.11	0.11	0.13
Disability type: other	0.06	0.06	0.06	0.06	0.07	0.06	0.06
Firm characteristics							
Firm size 500+	0.35	0.53	0.29	0.46	0.66	0.53	0.19
Self-insured firm	0.21	0.33	0.17	0.29	0.38	0.36	0.12
Observations	131,219	33,412	97,807	71,527	23,091	24,460	23,976

Note: Data provided by ODCBS. Column 1 consists of disabling claims between 2011 and 2017. Columns 2 and 3 distinguish claims that were accommodated and not accommodated, respectively. The analysis sample (column 4) consists of the subset of claims for which the firm had at least one claim between 2005–2009 and one claim between 2011–2017 (excluding claims in the first two quarters of 2013), and workers who worked at least 250 hours in the quarter prior to disability. Columns 5 through 7 distinguish claims in the analysis sample for which the firm's accommodation rate in 2005–2009 was in the top, middle, or bottom tercile, respectively. Prior quarterly earnings are from the quarter prior to the quarter of disability. Claim days are calendar days, while claim days with time loss paid are days in which time loss benefits were paid. Reported values are means, and standard deviations in parentheses.



(a) Separation rate (b) Quit rate

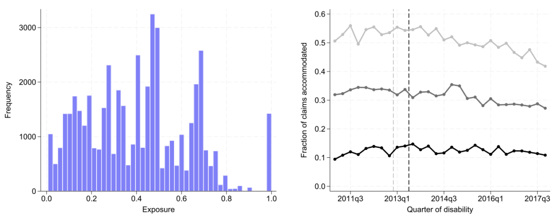
FIGURE 1.—Fraction of claims that use EAIP, by industrywide separation rates. *Notes*: Data from the ODCBS merged to two-digit industry month separation and quit rates from the Job Openings and Labor Turnover Survey. Sample consists of disabling claims between 2011 and 2017, aggregated to the two-digit industry level. Circles are proportional to the number of claims within that industry in the sample. Lines represent fitted regression lines through the aggregate data, taking out year-month effects, and shaded areas are 95% confidence intervals.

Read:

- Table I establishes the basic descriptive landscape. In the full 2011–2017 sample, 25% of claims use EAIP. In the analysis sample the rate is even higher, 31%.
- Accommodated claims are not obviously the "easy" cases. They have higher medical costs and longer durations on average, so positive selection on severity is plausible.
- Accommodation is far more common in large and self-insured firms. This is the first indication that firm incentives and insurance design matter.

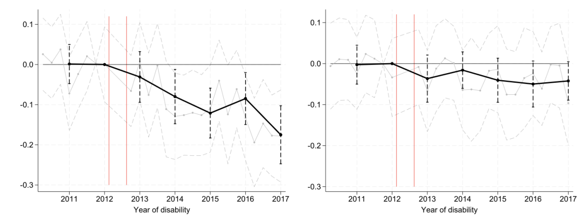
- Figure 1 is especially important for the model: accommodation rates are lower in industries with higher separation rates. This is exactly what one would expect if turnover weakens firms' incentives to invest in workers through accommodation.

5.2 Figure 2 and Figure 3: the exposure design and pretrends



(a) Distribution of exposure (b) Accommodation rate by exposure tercile

FIGURE 2.—Distribution of exposure and accommodation rates by exposure tercile. *Notes*: Data provided by ODCBS. Panel (a) reports the number of claims by exposure bin in our analysis sample, among claims in firms with positive exposure. 33% of the sample has zero exposure. Panel (b) reports the fraction of claims that are accommodated by quarter from 2011–2017, by exposure tercile (darkest solid line is highest exposure tercile, lightest solid line is lowest exposure tercile). Terciles are constructed from the continuous exposure measure. Dashed gray and black vertical lines delineate quarters before and after the date of the policy change announcement and implementation, respectively. For the sake of completeness, we also include claims from the first two quarters of 2013 claims in panel (b) even though they are excluded from the analytic sample.



(a) Accommodation (b) Employment four quarters later

FIGURE 3.—Dynamic effects of policy change on accommodation and employment. *Notes*: Data provided by ODCBS. Black (gray) dots denote the estimated annual (quarterly) coefficients on the interaction of exposure and year (quarter) from a regression of the outcome variable on exposure, quarter, and the interaction between exposure and year (2012 is omitted), along with a broad set of worker, firm, and disability controls. Estimate for 2013 excludes claims from the first two quarters of the year. Dashed black vertical lines report 95% confidence intervals for annual coefficients, dashed gray connected lines report 95% confidence intervals for quarterly coefficients. Vertical lines distinguish time periods before and after the policy announcement and implementation dates.

Read:

- Figure 2(a) shows that the exposure distribution is broad, so the continuous-treatment design has meaningful variation.
- Figure 2(b) shows similar trends in accommodation before the policy change and a noticeably larger post-policy decline for high-exposure firms. Low-exposure firms barely move.
- Figure 3 sharpens this point by plotting year- and quarter-specific interaction coefficients. The pre-2013 coefficients are close to zero, while the post-policy coefficients turn negative.
- Together, these figures provide the visual argument for the identifying parallel-trends assumption in the continuous-DID design.

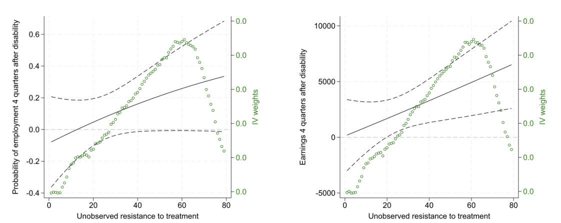
5.3 Table II and Figure 4: causal effects and treatment heterogeneity

TABLE II

DIFFERENCE-IN-DIFFERENCES ANALYSIS OF POLICY CHANGE ON ACCOMMODATION AND LABOR MARKET OUTCOMES FOUR QUARTERS AFTER DISABILITY.

		Four Quarters After Disability		
	EAIP	Employment	Earnings	New Firm
	(1)	(2)	(3)	(4)
Panel A. Difference-in-differences estimates				
Exposure × Post	−0.107	−0.035	−446.516	0.001
	(0.024)	(0.016)	(151.754)	(0.012)
Reweighted effect	−0.126	−0.032	−438	−0.001
Mean DV	0.312	0.721	7806.8	0.106
Observations	71,525	71,525	71,525	71,525
Panel B. IV estimates				
EAIP		0.329	4177.9	−0.0107
		(0.160)	(1624.1)	(0.108)
Mean DV		0.721	7806.8	0.106
Observations		71,525	71,525	71,525

Note: *Notes*: Data provided by ODCBS. The table reports β coefficients and standard errors (clustered by firm) from equation (1) in panel A and IV coefficients Ψ and standard errors (clustered by firm) from equation (3) in panel B. Panel A additionally reports average causal responses reweighted by the distribution of exposure in the sample. Dependent variables are whether the claim is accommodated through EAIP (column 1) and, four quarters after disability: whether the claimant is employed (column 2), the claimant's quarterly earnings (column 3), and whether the claimant is employed in a new firm (column 4). The Kleibergen–Paap F-statistic for the first stage of the IV is 19.1. All regressions include a broad set of worker, firm, and disability controls. Mean (SD) of exposure is 0.27 (0.27).



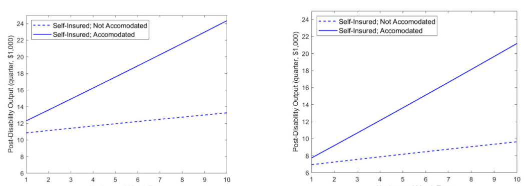
(a) Employment four quarters after disability (b) Earnings four quarters after disability

FIGURE 4.—Marginal treatment effects on employment and earnings. *Notes*: Data provided by ODCBS. Figures report marginal treatment effects (solid line) and 95% confidence intervals (dashed lines) on the left axis and the IV weights in circles on the right axis. Corresponding marginal treatment response curves are in Supplemental Figure F.4.

Read:

- Panel A of Table II shows that the policy change reduced accommodation, employment, and earnings in firms with higher exposure. The first-stage coefficient is -0.107, meaning a 10 percentage point increase in exposure implies about a 1.07 percentage point larger accommodation decline after the subsidy cut.
- At average exposure, this corresponds to a 2.9 percentage point decline in accommodation, the headline reduced-form effect.
- The IV estimates in panel B are much larger: accommodation raises employment by 0.329 and quarterly earnings by about $4,178. These are large local effects.
- There is no statistically meaningful effect on moving to a new firm. This is crucial because it supports the paper's interpretation that accommodation primarily raises general human capital rather than firm-specific attachment.
- Figure 4 shows negative selection on gains. The workers who are hardest to treat are the ones with the highest treatment effects. This explains why the IV estimate is so large: the policy instrument shifts accommodation exactly for those workers with large latent gains.

5.4 Figure 5 and Table III: productivity and the cost of accommodation



(a) High-skilled workers (b) Low-skilled workers

FIGURE 5.—Post-disability output of workers in self-insured and median exposure firms. *Notes*: The left (right) figures report the post-disability output of high-skilled (low-skilled) workers in self-insured and median exposure firms, by accommodation status. The x-axis in each figure denotes the worker-firm match's unobserved type decile.

TABLE III

PARAMETERS ESTIMATED WITHIN THE MODEL.

Param.	Description	Estimate	SE	Param.	Description	Estimate	SE
Net output, without disability: $f_{1,z,0}$				Net output, with disability: $f_{1,z,\xi}$			
$\alpha^0_{f,0}$	Baseline	14.144	(0.025)	$\alpha^0_{f,1}$	Baseline	−17.211	(0.055)
$\alpha^1_{f,0}$	Self-insured	1.835	(0.023)	$\alpha^1_{f,1}$	Self-insured	−1.485	(0.013)
$\alpha^2_{f,0}$	High skilled	0.103	(0.024)	$\alpha^2_{f,1}$	High skilled	1.171	(0.009)
$\alpha^3_{f,0}$	Exposure	0.023	(0.016)	$\alpha^3_{f,1}$	Exposure	1.266	(0.002)
				$\alpha^4_{f,1}$	Unobs. type	−0.013	(0.007)
Accommodation cost shock std. dev.: $\sigma_{\xi,z}$							
$\alpha_{\sigma,0}$	Baseline	4.762	(0.026)				
$\alpha_{\sigma,1}$	Unobs. type	−0.631	(0.008)				

Note: Output is quarterly and is expressed in units of $1000. The parameters and their functional forms are introduced in equations (17) and (18). Standard errors of estimates are in parentheses.

Read:

- Figure 5 shows that accommodated workers have much higher post-disability output than nonaccommodated workers, especially at high unobserved match types. The gain is particularly steep for low-skilled workers.
- Table III is central. Baseline quarterly output without disability is 14.144 (in $1,000s), while baseline net output during disability is *negative*, at -17.211 (again in $1,000s).
- This means accommodation is a costly investment during the disability spell, not an immediate profit opportunity. Firms incur short-run losses in exchange for future gains.
- The accommodation cost shock standard deviation is 4.762, and it declines with the unobserved type ($\alpha_1^\sigma = -0.631$), implying that high-return types both gain more from accommodation and respond more strongly to subsidies.

5.5 Table IV and Figure 6: model fit and mechanisms

Read:

TABLE IV

WITHIN-SAMPLE FIT OF TARGETED MOMENTS.

	Accommodation Rate		Wages in Period 1	
	Data	Model	Data	Model
DID coefficient	−0.11	−0.07		
Levels by subsample:				
Self-insured	0.43	0.50	11.24	11.20
Not self-insured	0.27	0.23	9.22	9.80
High skilled	0.34	0.38	13.46	12.19
Low skilled	0.28	0.23	6.36	8.22
High exposure	0.46	0.39	10.63	10.47
Low exposure	0.16	0.17	8.97	9.81
Prepolicy change	0.32	0.33	9.64	10.15
	Post-Disability Employment		**Post-Disability Wages**	
	Data	Model	Data	Model
DID coefficient	−0.04	−0.03	−0.45	−0.56
Overall level	0.72	0.77	9.19	8.70

Note: Wages are quarterly and are expressed in units of $1000.

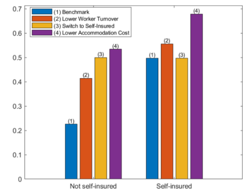
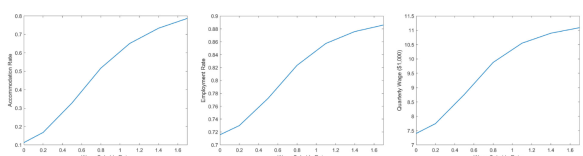


FIGURE 6.—Comparative statics of accommodation decisions. *Notes*: The figure shows changes in accommodation rates in four scenarios: (i) the benchmark (i.e., the outcome under the estimated samples; (ii) the effect of lowering worker's job-to-job transition rate λ_z to 25% of its estimated value; (iii) the effect of switching to the self-insured contract; (iv) and the effect of increasing the net output of workers with disabilities by 25% of the absolute value of $a^0_{f,1}$.

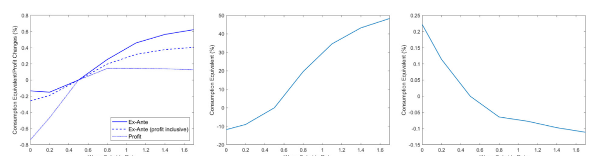
- Table IV shows that the structural model broadly matches the targeted moments. For example, the model reproduces the accommodation DID coefficient reasonably well (−0.07 in the model versus −0.11 in the data) and also tracks the employment and wage DID moments.
- Figure 6 is the mechanism figure. Lower worker turnover raises accommodation rates, especially in imperfectly experience-rated firms.
- Moving imperfectly experience-rated firms to self-insured status raises accommodation substantially, confirming the importance of the fiscal externality through workers' compensation premiums.
- Lowering accommodation costs also raises accommodation, which confirms that not all underaccommodation comes from finance or turnover frictions; there is also a large static cost side.

5.6 Figure 7 and Figure 8: counterfactual labor-market and welfare effects



(a) Accommodation rate (b) Post-disability employment (c) Post-disability wage

FIGURE 7.—Labor market impacts of wage subsidies for accommodation. *Notes*: Each subfigure reports outcomes from counterfactual experiments in which we vary the wage subsidy rate for accommodation. The benchmark case is a wage subsidy rate of 50%.

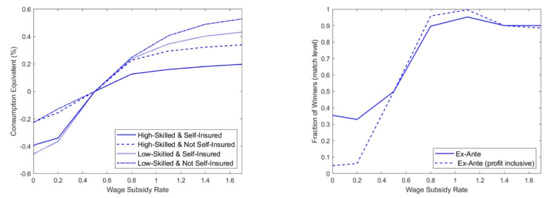


(a) Ex ante (b) Ex post (disabled) (c) Ex post (nondisabled)

FIGURE 8.—Average welfare impacts of wage subsidies for accommodation. *Notes*: The figure shows the welfare and profit impacts of counterfactual wage subsidy rates relative to a 50% wage subsidy. The left subfigure includes three measures: (i) the change in average ex ante worker welfare; (ii) the change in average ex ante worker welfare inclusive of firm profit as an ex post transfer to workers; and (iii) the change in average ex ante firm profit. The middle subfigure reports the change in average (ex post) welfare for workers experiencing disability shocks. The right subfigure reports the change in average (ex post) welfare for workers not experiencing disability shocks.

Read:

- Figure 7 shows that the subsidy rate matters enormously. Eliminating the wage subsidy drives accommodation down from 33% to 11%.
- The same policy lowers post-disability employment by 7% and quarterly wages by 15% (about $1,358). So the accommodation margin is quantitatively first order.
- Figure 8 shows that ex ante average welfare effects are positive but modest. Relative to the 50% benchmark, even very high subsidies raise average worker welfare by at most about 0.6% in consumption-equivalent terms, while eliminating subsidies lowers it by about 0.3%.



(a) Ex-ante welfare by worker-firm match

(b) Fraction of winners

FIGURE 9.—Distributional implications of wage subsidies for accommodation. *Notes*: The left subfigure shows the welfare impact of wage subsidies by the type of worker-firm match, measured by the change in average ex ante worker welfare inclusive of firm profit as an ex post transfer to workers relative to a 50% wage subsidy. The right subfigure shows distributional outcomes from counterfactual wage subsidy rates relative to a 50% wage subsidy. We report the fraction of matches that experience strictly positive welfare gains relative to the benchmark based on two welfare measures: ex ante worker welfare and ex ante worker welfare inclusive of firm profit as an ex post transfer to workers.

- These average effects are modest because disability is infrequent and because nonaccommodated workers still receive time-loss benefits.
- But the ex post welfare effects for workers who actually experience disability are large: eliminating subsidies reduces their welfare by about 10%, while raising the subsidy to 100% raises it by about 30%.

5.7 Figure 9: who benefits from more generous subsidies?

Read:

- Figure 9 shows that the welfare gains from subsidies are larger for low-skilled workers and for workers in imperfectly experience-rated firms.
- This means accommodation subsidies are not just efficiency enhancing; they are also redistributive toward groups facing larger underaccommodation.
- The right panel shows that if subsidies were eliminated, more than 90% of worker-firm matches would experience welfare losses relative to the current 50% benchmark.
- This is strong evidence that the benchmark subsidy is not too generous from the standpoint of broad ex ante welfare.

6 Short summary

- **Method.** The paper combines an exposure-based policy-change design with a dynamic bargaining model of workplace disability, accommodation, and workers' compensation.
- **Identification.** A 2013 subsidy cut interacted with firms' historical accommodation exposure identifies the causal effect of accommodation. An MTE framework then recovers heterogeneity in gains, and indirect inference identifies the structural parameters.
- **Application.** In Oregon's workers' compensation system, accommodation is common but far from universal, especially in firms with high turnover and weak experience rating.
- **Main result.** Accommodation substantially improves long-run employment and earnings, while worker turnover and imperfect experience rating lead firms to underaccommodate. Wage subsidies for accommodation improve labor-market outcomes and welfare, especially for low-skilled workers.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that firm accommodation after workplace disability responds strongly to financial incentives.
- It also establishes that accommodation has large causal effects on subsequent employment and earnings.
- In the structural model, underaccommodation arises because firms do not fully capture either the future labor-market surplus or the insurance cost savings generated by accommodation.

7.2 Economic intuition

- Accommodation is costly in the short run because the worker is less productive during recovery and the firm may have to rearrange tasks or absorb other costs.
- But accommodation can preserve or build human capital, making the worker more productive and more employable later.
- If workers may leave, part of that future gain accrues to another employer. If insurance is only partially experience rated, part of the financial savings accrues to the insurance pool. In both cases, the original firm invests too little.

7.3 Takeaway

This paper shows that the design of disability-related social insurance cannot ignore firms. Accommodation subsidies change firm behavior, improve workers' long-run outcomes, and can raise welfare even after accounting for the tax cost of financing them. In the Oregon setting, employer subsidies are not a side feature of the system — they are a core policy instrument for correcting underinvestment in post-disability labor-market recovery.